

DHEERAJ KUMAR

+91-6299848516 | dheeraj116232@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [LeetCode](#) | [Code360](#)

Education

National Institute of Technology (NIT), Tiruchirappalli
B.Tech in Civil Engineering (Minor in Computer Science)

2022 – 2026
CGPA: **7.64**

Technical Skills

Languages	Python, JavaScript, SQL, C/C++
Frontend	React.js, HTML5, CSS3, Tailwind CSS
Backend	FastAPI, Node.js, REST APIs
AI / ML	LangChain, LangGraph, RAG, LLM APIs, Agentic AI, Scikit-learn, XGBoost, TensorFlow, PCA, K-Means, Feature Engineering
Data & Analytics	Pandas, NumPy, Power BI, EDA, A/B Testing, Cohort/Funnel Analysis
Databases & Cloud	PostgreSQL, MongoDB, MySQL, AWS Bedrock, Git, Vercel, Render

Experience

Data Science Research Intern

May 2025 – Jul 2025

NIT Trichy

- Conducted **EDA** on **multi-million-record banking datasets** using Python, SQL, Pandas, and Seaborn, uncovering insights that improved **decision-making efficiency by 25%**.
- Applied **K-Means** clustering and **PCA** to segment **100K+** FMCG customers, increasing campaign targeting precision and **ROI by 30%**.
- Engineered a **RAG chatbot** using **LangChain**, **LLMs**, and **MongoDB**, automating Annual Report analysis and reducing **manual effort by 40%**.

Full-Stack Web Developer Intern

Jun 2025 – Aug 2025

NIT Trichy

- Designed and developed **10+ responsive web pages** (CV, awards, publications) for a faculty member using **React.js**, streamlining content presentation and navigation.
- Implemented a **file management system** using **Node.js** and **REST APIs**, reducing content update time from **1 day to 5 minutes**.
- Owned the project **end-to-end** – UI design, backend integration, and deployment – delivering a fully functional, self-service platform within **2 months**.

Projects

AI Business Intelligence Platform

[GitHub](#) | [Live](#)

Python, LangGraph, FastAPI, Next.js, PostgreSQL, Docker

- Engineered a **5-agent LangGraph workflow** for automated data cleaning, KPI analytics, forecasting, anomaly detection, and **LLM-powered** executive reporting.
- Reduced business analysis time by **90%+ (30 sec)**, processing **10+ file formats** and generating **CEO-ready PDF reports**.

Multi-Agent Travel AI – Automated Travel Planning

[GitHub](#) | [Live](#)

Python, LangGraph, LangChain, Streamlit, PostgreSQL, Groq, Docker

- Built a **4-agent AI pipeline** (Flight → Hotel → Itinerary → Summary) generating complete travel plans with **live API integration** and cost optimization.
- Integrated AviationStack, Tavily, PostgreSQL, and LangGraph checkpointing, delivering **downloadable Markdown/PDF itineraries** in **10–15 sec**.

AI Blog Generation Platform

[GitHub](#) | [Live](#)

Python, LangGraph, AWS Bedrock, Streamlit, Tavily

- Developed a **4-stage multi-agent pipeline** (Plan → Research → Write → Image Generation) supporting Grok, Tavily, and automated web research.
- Generated **SEO-optimized blogs** with **Markdown/PDF/ZIP export**, persistent storage, and deployment on Render.

Layoff Risk Prediction

[GitHub](#)

Python, Scikit-learn, Pandas, PCA

- Built ML models (Logistic Regression, Random Forest, Gradient Boosting) predicting layoff risk with **79% accuracy** using **PCA** and **k-fold cross-validation**.
- Improved prediction reliability through **feature engineering**, preprocessing, and **model optimization**.

Achievements & Certifications

- [Oracle AI Foundations Associate](#) | [Oracle Generative AI Professional](#)
- Naukri Scholarship 2025: College Rank #2, National Rank #614 among **25,000+** engineering students (IIT/NIT/IIIT/BITS)
- HCLTech Hackathon 2026: Built data-analytics solutions for real-world business challenges (3-sprint MVP)
- National Mathematics Olympiad (NMO): **AIR 513**
- 800+** DSA problems solved (LeetCode & Code360)